

Coarse and Fine-Grained Dog Breed Discovery Using Pretrained CNN Feature Extractors

Salamot Fakoya Isabella Ran

University of Texas at Dallas

Erik Jonsson School of Engineering and Computer Science

{sif240000, ixr220014}@utdallas.edu

Abstract

This project investigates coarse and fine-grained image classification using pretrained convolutional neural networks as unsupervised feature extractors. We evaluate ResNet-18 and VGG-16 on CIFAR-10 (coarse dog labels) and Imagewoof (10 fine-grained dog breeds), applying K-Means clustering to discover breed groupings without any breed labels at clustering time. We further conduct three robustness experiments on the Oxford-IIIT Pet dataset: SAM-based background removal, controlled lighting degradation, and an architecture comparison extending to Vision Transformers (ViT-B/16, ViT-L/16). Image resolution is the primary bottleneck; SAM background removal consistently degrades CNN performance; lighting affects dog clustering more than cats; and ViT substantially outperforms both CNNs. ResNet-18 achieves ARI 0.581 on Imagewoof (purity 77.3%) and ARI 0.530 on Oxford Dogs (purity 90.2%). VGG-16 achieves ARI 0.571 on Imagewoof and ARI 0.493 on Oxford Dogs. ViT-B/16 reaches ARI 0.796 and NMI 0.925 on Oxford Dogs, a 62% improvement over VGG-16.

1. Introduction

Domestic dogs have been selectively bred over thousands of years, producing remarkable phenotypic diversity across hundreds of recognized breeds. Recognizing individual breeds is a challenging computer vision problem: inter-class variation between breeds can be subtle while intra-class variation due to pose, lighting, and background is large. This project investigates whether pretrained CNN models can discover breed-level structure in an entirely unsupervised manner, without any breed labels at clustering time.

We evaluate this pipeline across two scales of difficulty. At the coarse level, CIFAR-10 provides a single undifferentiated dog class at 32×32 pixels. At the fine-grained

level, Imagewoof provides 10 visually similar breeds at higher resolution. We further extend to Oxford Dogs (25 breeds) and Oxford Cats (12 breeds) for robustness experiments. The key contributions are: (1) an empirical analysis of ResNet-18 and VGG-16 for fine-grained unsupervised breed clustering; (2) a SAM-based background removal study across three datasets and both models; (3) a lighting robustness analysis across dog and cat species; and (4) an architecture comparison demonstrating that Vision Transformers overcome the background averaging limitation of CNNs.

1.1. Related Work

Rhodes [11] achieved $\approx 67\%$ top-1 accuracy on 133 dog breeds using facial keypoint localization and normalization, highlighting that fine-grained classification requires part-level discriminative features. ResNet [1] and VGG-16 [2] are used here as frozen feature extractors producing 512-d and 4,096-d representations respectively. ViT [3] replaces convolutional pooling with self-attention, selectively weighting discriminative image regions. SAM [4] provides promptable segmentation used in our background removal experiments. The Oxford-IIIT Pet dataset [5] provides independent evaluation benchmarks sourced from Flickr, independent of ImageNet [6].

2. Proposed Approach

The pipeline is: raw image $\rightarrow$ frozen pretrained feature extractor $\rightarrow$ StandardScaler normalization $\rightarrow$ PCA $\rightarrow$ K-Means $\rightarrow$ evaluation against ground-truth breed labels. No breed labels are used during clustering; labels are used only at evaluation time.

ResNet-18. 18-layer residual network [1] with AdaptiveAvgPool output. Classification head removed; produces 512-d feature vectors (11.7M parameters).

VGG-16. 13 convolutional layers plus three FC layers [2]. Final classification layer (fc8) removed; fc6 and fc7 retained, producing 4,096-d feature vectors (134.3M parameters, loaded with `VGG16_Weights.IMAGENET1K_V1`).

ViT models (Experiment 3). ViT-B/16 (768-d class token, 86M parameters) and ViT-L/16 (1,024-d, 307M parameters) [3]. Self-attention weights discriminative foreground regions without explicit segmentation.

Clustering and evaluation. K-Means with k-means++ initialization, 10 restarts, 300 maximum iterations. Optimal K selected by maximizing silhouette score on a 3,000-point training subsample. Adjusted Rand Index (ARI) is the primary metric. Normalized Mutual Information (NMI) and average cluster purity are reported alongside.

3. Experiments

3.1. Empirical Settings

Datasets. *CIFAR-10 Dogs:* 6,000 images (4,500/500/1,000) at 32×32 , upsampled to 224×224 . No breed sub-labels; abandoned after resolution failure. *Imagewoof:* 9,025 images (7,220/902/903) across 10 dog breeds (class sizes 580–949), resized to 224×224 , 80/10/10 split, seed 42. *Oxford Dogs:* 4,978 images (3,982/497/498) across 25 breeds, Flickr-sourced, independent of ImageNet. *Oxford Cats:* 2,371 images (1,896/237/238) across 12 breeds, same source.

Preprocessing. All images resized to 224×224 with $[0.5, 0.5, 0.5]$ normalization. Post-extraction features standardized with `StandardScaler` (fit on training only). PCA fitted on training features and applied to all splits [8].

Hardware and hyperparameters. All models frozen (`eval() + torch.no_grad()`) with no optimizer or training epochs [9]. Experiments ran on Tesla T4 GPU (PyTorch 2.10.0+cu128). VGG-16 SAM extraction took 74.3 min (per-image SAM inference); batched baseline extraction ran at ~ 68 it/s. K sweeps: $\{5, 8, 10, 12, 15, 20\}$ for Imagewoof/Oxford Cats; $\{5, 8, 10, 12, 15, 20, 25\}$ for Oxford Dogs; $\{2, \dots, 10\}$ for CIFAR-10. t-SNE [10] (perplexity 30, seed 42) used for visualization only.

Evaluation metrics. *Silhouette score:* intrinsic geometric measure $[-1, 1]$, used for K selection. Typically low for high-dimensional image features even when clustering is semantically strong. *ARI:* primary metric, cluster agreement with true labels adjusted for chance, $[0, 1]$. *NMI:*

shared information between predicted clusters and true labels, $[0, 1]$. ARI/NMI not computable for CIFAR-10 (no breed labels).

3.2. Empirical Results

3.2.1 CIFAR-10 — Failure Case

Both models fail on CIFAR-10 dogs. ResNet-18 peaked at silhouette 0.014 (K=5) with five imbalanced clusters (30.1%/21.0%/20.0%/ 16.9%/12.0%). VGG-16 peaked at silhouette 0.054 (K=2) with clusters of 79.6% and 20.4%. Native 32×32 resolution destroys breed discriminative spatial detail; upscaling to 224×224 recovers none of it, and the absence of breed sublabels prevents ARI/NMI evaluation. Both issues motivated transitioning to higher-resolution datasets.

3.2.2 Baseline Clustering — Imagewoof

Tab. 1 reports results. Fig. 1 and Fig. 2 show t-SNE projections of the Imagewoof feature space for ResNet-18 and VGG-16 respectively. In both cases the left panel (true breed labels) and right panel (K-Means cluster assignments) show strong spatial alignment, confirming that both models recover breed level structure without label supervision.

ResNet-18 achieves ARI 0.581, NMI 0.687, silhouette 0.066, and correctly selects K=10 matching the true breed count. Per-breed purity averages 77.3%, ranging from 68.2% (Shih-Tzu) to 84.5% (Rhodesian Ridgeback and Shiba Inu). Only one cluster is shared: Beagle and English Foxhound in Cluster 3 which is a biologically meaningful confusion between two visually similar hounds. The confusion matrix in Fig. 3 shows strong diagonal dominance confirming per-breed cluster ownership.

VGG-16 achieves an ARI 0.571, NMI 0.680, silhouette 0.056, selecting K=12. Per-breed purity averages 70.8%, ranging from 55.5% (English Foxhound) to 77.3% (Old English Sheepdog). The confusion matrix in Fig. 4 shows every breed with its own dominant cluster and no shared assignments. ResNet-18 slightly outperforms VGG-16 on Imagewoof despite having $8\times$ fewer feature dimensions, suggesting global average pooling is the shared bottleneck rather than feature capacity.

3.2.3 Baseline Clustering — Oxford Pets

Fig. 5 and Fig. 6 show t-SNE projections for ResNet-18 on Oxford Dogs and Oxford Cats respectively. On Oxford Dogs the 25 breeds form clearly separated compact blobs, explaining the high purity of 90.2%. On Oxford Cats the 12 breeds show substantially more inter-cluster overlap, consistent with the weaker ARI of 0.261.

On Oxford Dogs, ResNet-18 achieves an ARI 0.530, NMI 0.789, and average purity 90.2%. Of 25 breeds, 22 ex-

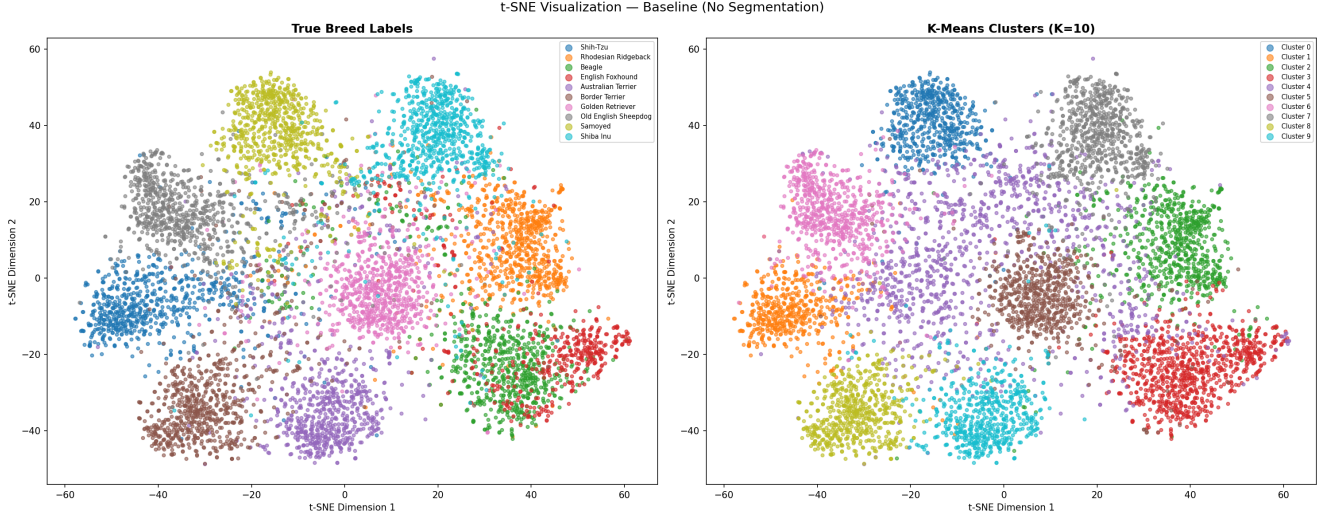


Figure 1. t-SNE visualization of ResNet-18 features on Imagewoof (K=10). Left: true breed labels. Right: K-Means cluster assignments. Strong spatial alignment between panels confirms breed-level structure is recoverable without label supervision. ARI 0.581.

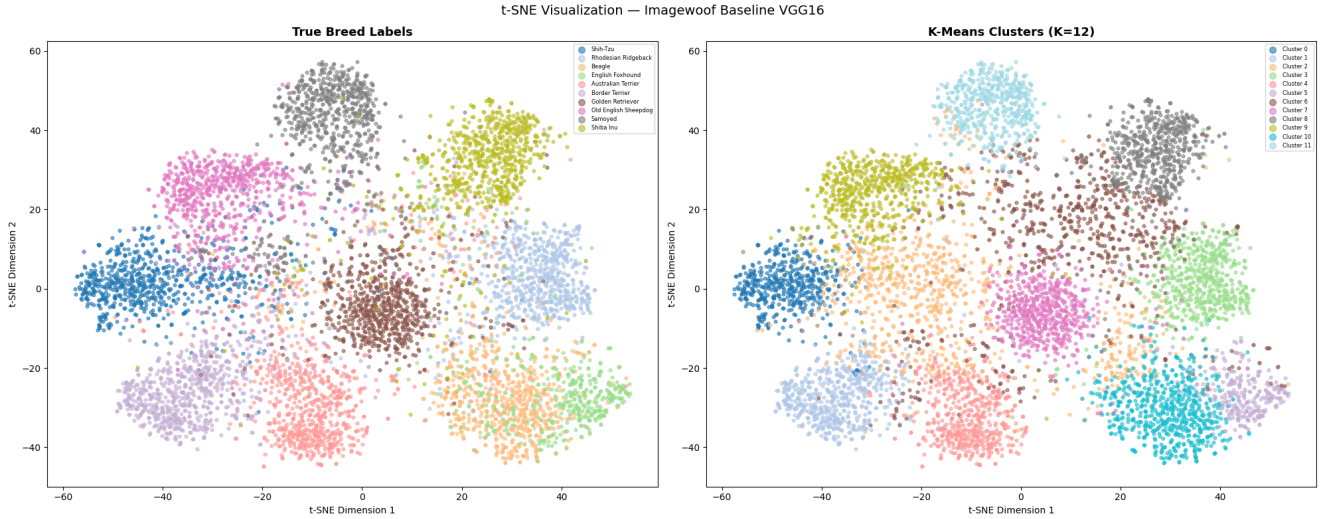


Figure 2. t-SNE visualization of VGG-16 features on Imagewoof (K=12). Left: true breed labels. Right: K-Means cluster assignments. Breed regions are broadly separable; Samoyed (upper left) and Shiba Inu (upper right) form the tightest isolated clusters. ARI 0.571.

ceed 80% purity and 18 exceed 90%. Cluster sharing is biologically meaningful throughout: Basset Hound and Beagle share Cluster 7; Boxer and Pug share Cluster 5; Great Pyrenees and Samoyed share Cluster 13; Leonberger and Newfoundland share Cluster 2. VGG-16 achieves an ARI 0.493 and a NMI 0.765, below ResNet-18 on dogs.

On Oxford Cats, VGG-16 marginally outperforms ResNet-18 (ARI 0.283 vs. 0.261). However the VGG-16 silhouette at K=5 is effectively zero and becomes negative at all higher K values, indicating near-degenerate cluster geometry for cats. The species asymmetry, dogs more separable than cats, persists across all models and experiments.

3.2.4 Experiment 1 — SAM Background Removal

SAM [4] was used to replace image backgrounds with black (pixel value 0) or white (pixel value 1) fill, using a five-point multiprompt strategy with coverage filter 10%–80%. Fig. 7 shows the ARI and NMI summary across all datasets and pipelines. Tab. 2 reports the full numerical results for both models.

The baseline outperforms both SAM variants for both models across all 18 comparisons without exception. Two causes explain the degradation. First, imperfect masks corrupt features for off center or unusually posed animals. Second, flat fills disrupt the feature statistics that ImageNet-

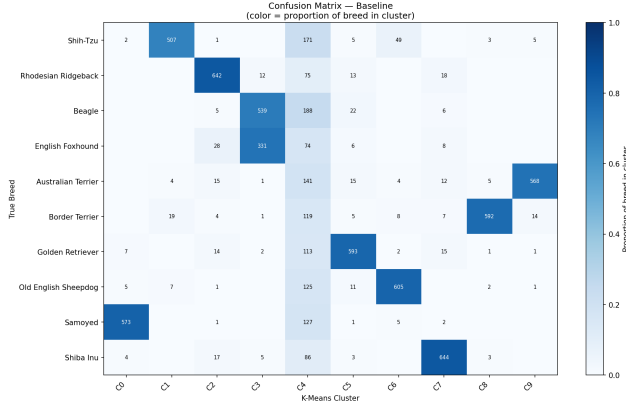


Figure 3. Confusion matrix — ResNet-18 Imagewoof baseline. Color encodes proportion of each breed in its dominant cluster. Strong diagonal dominance; only Beagle/English Foxhound share a cluster (Cluster 3).

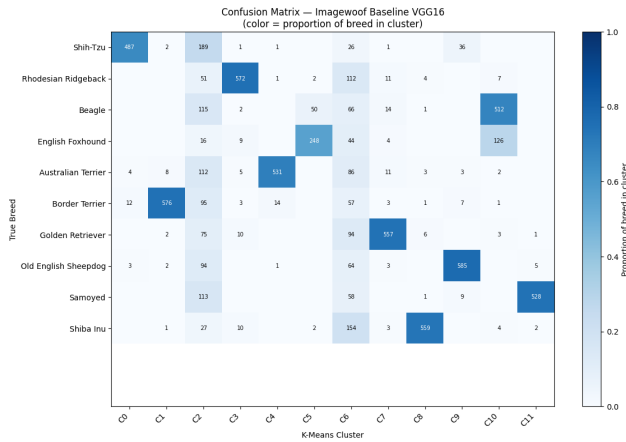


Figure 4. Confusion matrix — VGG-16 Imagewoof baseline (K=12). Every breed occupies its own dominant cluster with no sharing. English Foxhound (55.5%) and Shih-Tzu (65.5%) show the most diffuse assignments; Old English Sheepdog (77.3%) the tightest.

pretrained networks learned to interpret — neither model was ever trained on images with artificial flat backgrounds. A color specific effect also applies: white fill degrades light coated breeds (Samoyed, Shih-Tzu) because the animal’s coat becomes indistinguishable from the background at the pixel level.

3.2.5 Experiment 2 — Controlled Lighting

Tab. 3 shows both models under brightness factors 1.0 (normal), 0.4 (dark, −60%), and 0.2 (very dark, −80%) applied via torchvision ColorJitter.

Both models show progressive degradation with reduced brightness. For dogs, ResNet-18 ARI drops 52% at dark

Dataset	Model	K	ARI	NMI	Sil.
CIFAR-10	ResNet-18	5	N/A	N/A	0.014
CIFAR-10	VGG-16	2	N/A	N/A	0.054
Imagewoof	ResNet-18	10	0.581	0.687	0.066
Imagewoof	VGG-16	12	0.571	0.680	0.056
Oxford Dogs	ResNet-18	15	0.530	0.789	0.063
Oxford Dogs	VGG-16	15	0.493	0.765	0.051
Oxford Cats	ResNet-18	5	0.261	0.482	0.036
Oxford Cats	VGG-16	5	0.283	0.533	0.000

Table 1. Baseline clustering results (validation set). ResNet-18 leads on dogs; VGG-16 edges ahead on cats. Neither model consistently dominates.

Dataset	Model	Pipeline	ARI	NMI	Purity
Imagewoof	ResNet	Baseline	0.580	0.690	77.3%
	ResNet	SAM Black	0.251	0.411	46.6%
	ResNet	SAM White	0.143	0.254	50.6%
	VGG-16	Baseline	0.571	0.680	70.8%
	VGG-16	SAM Black	0.171	0.310	~45%
	VGG-16	SAM White	0.132	0.251	~45%
Ox. Dogs	ResNet	Baseline	0.530	0.789	90.2%
	ResNet	SAM Black	0.248	0.539	68.1%
	ResNet	SAM White	0.362	0.614	65.1%
	VGG-16	Baseline	0.493	0.765	85.6%
	VGG-16	SAM Black	0.206	0.518	68.4%
	VGG-16	SAM White	0.318	0.608	68.2%
Ox. Cats	ResNet	Baseline	0.261	0.482	73.0%
	ResNet	SAM Black	0.143	0.316	58.2%
	ResNet	SAM White	0.261	0.451	54.2%
	VGG-16	Baseline	0.283	0.533	85.8%
	VGG-16	SAM Black	0.152	0.325	60.4%
	VGG-16	SAM White	0.194	0.402	64.8%

Table 2. Experiment 1: SAM background removal. Baseline outperforms both SAM variants for both models across all 18 comparisons.

and 77% at very dark; VGG-16 drops 40% at dark and 85% at very dark. For cats, degradation is more moderate: ResNet-18 drops 23% and 69%; VGG-16 drops 42% and 67%. The dog-cat asymmetry reflects the nature of discriminative features — dog breeds are largely differentiated by coat color and markings (more brightness dependent), while cat breeds rely more on body shape and broad coat texture (more brightness robust). Both models reach similar very dark ARI levels on dogs (~0.07–0.12), confirming both rely on the same color-dependent feature types.

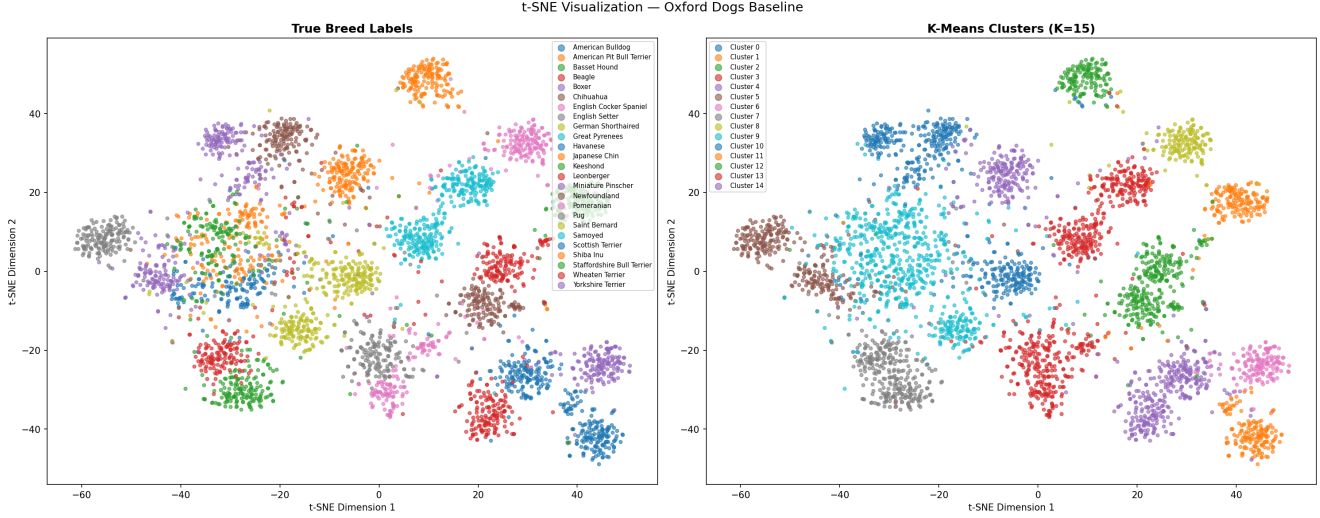


Figure 5. t-SNE visualization — ResNet-18 Oxford Dogs baseline (K=15, 25 breeds). Left: true breed labels showing tight, well-separated clusters. Right: K-Means assignments. Breeds with extreme phenotypes (Pomeranian, Pug, Saint Bernard) form isolated compact regions. ARI 0.530, purity 90.2%.

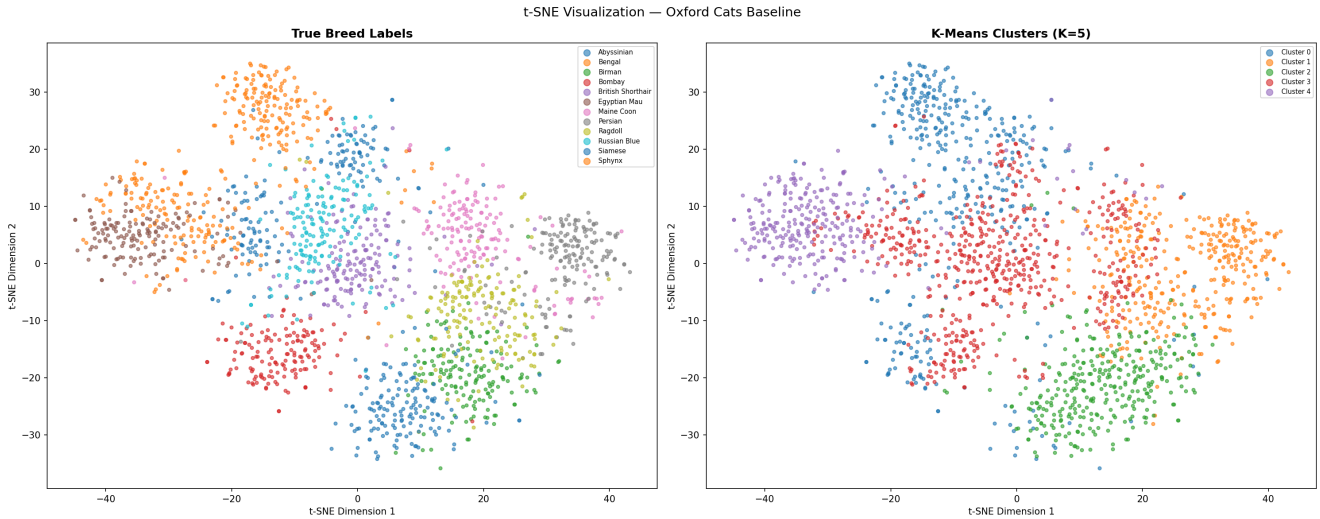


Figure 6. t-SNE visualization — ResNet-18 Oxford Cats baseline (K=5, 12 breeds). Left: true breed labels showing considerably more inter-class overlap than dogs. Right: K-Means assignments. Persian (top) and Sphynx (far left) are visually distinctive and well separated; most other breeds overlap substantially. ARI 0.261.

3.2.6 Experiment 3 — Architecture Comparison

Fig. 8 shows the full architecture comparison including all four models on Oxford Dogs and Cats. Tab. 4 reports the numerical results. Neither CNN consistently outperforms the other: ResNet-18 leads on dogs (0.530 vs. 0.493) while VGG-16 leads on cats (0.283 vs. 0.261), confirming that dimensionality alone does not drive clustering quality.

Both ViT models dramatically outperform both CNNs. ViT-B/16 achieves ARI 0.796 on Oxford Dogs, representing a 50% improvement over ResNet-18 and 62% over

VGG-16. ViT silhouette scores also improve markedly: ResNet-18 0.063 vs. ViT-B/16 0.191 on dogs, confirming that ViT features form geometrically tighter clusters in addition to being semantically stronger. ViT-B/16 slightly outperforms ViT-L/16 on dogs (0.796 vs. 0.773), as both share the same 90 ImageNet dog breed classes during pretraining. ViT-L/16 substantially outperforms ViT-B/16 on cats (0.437 vs. 0.315), where subtle cat breed distinctions benefit from higher representational capacity.

Both CNNs use global average pooling which averages background and foreground pixels equally, diluting breed-

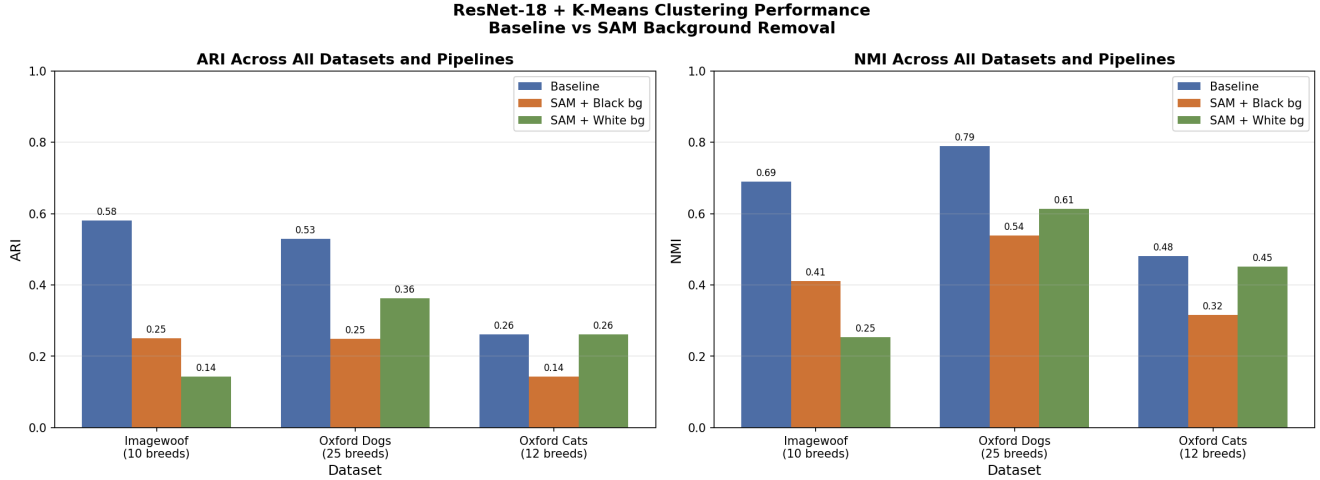


Figure 7. Experiment 1: ResNet-18 ARI (left) and NMI (right) across all datasets and pipeline variants. The baseline (blue) outperforms both SAM variants (orange, green) in all nine comparisons. Degradation is most severe on Imagewoof where white fill reduces ARI from 0.58 to 0.14.

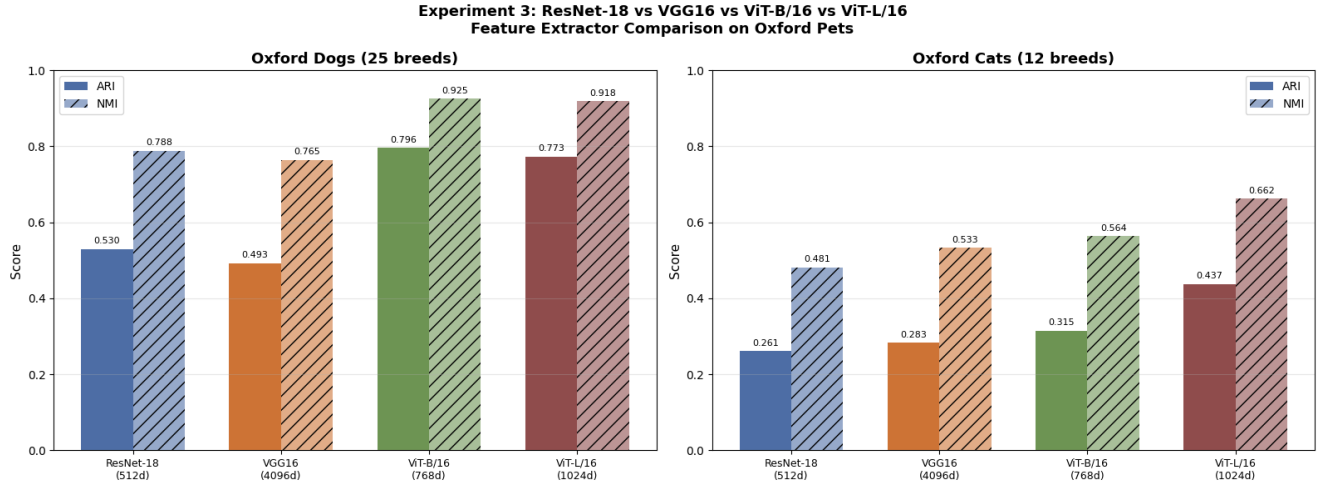


Figure 8. Experiment 3: Architecture comparison — ResNet-18, VGG-16, ViT-B/16, and ViT-L/16 on Oxford Dogs (left) and Oxford Cats (right). Solid bars = ARI (primary metric), hatched bars = NMI. ViT-B/16 achieves ARI 0.796 on Oxford Dogs, a 62% improvement over VGG-16 (0.493) and 50% over ResNet-18 (0.530).

discriminative features. ViT self-attention implicitly focuses on discriminative regions, solving the same problem that SAM preprocessing failed to address without any segmentation overhead.

4. Discussion

Resolution is the primary gating factor. CIFAR-10 at 32×32 fails completely for both CNNs. Imagewoof at 160px enables ARI above 0.57 for both models. Image resolution is a hard prerequisite for fine-grained unsupervised clustering that supersedes all other design choices.

VGG-16 vs. ResNet-18. The two CNNs perform comparably with no consistent winner. ResNet-18 leads on Imagewoof (0.581 vs. 0.571) and Oxford Dogs (0.530 vs. 0.493); VGG-16 leads on Oxford Cats (0.283 vs. 0.261). The $8 \times$ larger VGG-16 feature space does not translate to reliable gains, indicating that global average pooling — shared by both — is the binding constraint, not feature dimensionality.

SAM fails for both models. SAM degrades clustering across all 18 dataset-pipeline comparisons for both ResNet-18 and VGG-16. The failure is model agnostic: flat artificial fills are incompatible with ImageNet’s pretrained fea-

Model	Brightness	Species	ARI	NMI	Drop
ResNet-18	Normal	Dogs	0.530	0.789	—
	Dark	Dogs	0.254	0.556	−0.276
	V.Dark	Dogs	0.124	0.377	−0.406
	Normal	Cats	0.261	0.482	—
	Dark	Cats	0.202	0.404	−0.059
	V.Dark	Cats	0.080	0.228	−0.181
VGG-16	Normal	Dogs	0.493	0.765	—
	Dark	Dogs	0.295	0.620	−0.198
	V.Dark	Dogs	0.075	0.266	−0.418
	Normal	Cats	0.283	0.533	—
	Dark	Cats	0.163	0.381	−0.120
	V.Dark	Cats	0.094	0.255	−0.189

Table 3. Experiment 2: Controlled lighting (validation set). Both models degrade substantially; dogs are affected more severely than cats at all brightness levels for both models.

Model	Dims	Species	ARI	NMI
ResNet-18	512	Dogs	0.530	0.789
VGG-16	4096	Dogs	0.493	0.765
ViT-B/16	768	Dogs	0.796	0.925
ViT-L/16	1024	Dogs	0.773	0.918
ResNet-18	512	Cats	0.261	0.482
VGG-16	4096	Cats	0.283	0.533
ViT-B/16	768	Cats	0.315	0.564
ViT-L/16	1024	Cats	0.437	0.662

Table 4. Experiment 3: Architecture comparison on Oxford Pets. ViT substantially outperforms both CNNs. VGG-16 and ResNet-18 show no consistent dominance across species.

ture statistics regardless of architecture.

CNN vs. ViT. ViT’s 62% relative improvement over VGG-16 on Oxford Dogs is the strongest evidence that architectural design, not feature dimensionality, determines fine-grained clustering capability. Self attention solves the background averaging problem implicitly.

Species asymmetry. Cat clustering is consistently weaker than dog clustering for all models. Dog breeds show more extreme phenotypic variation. ImageNet contains 90 dog breed classes and no cat breed classes, directly advantaging dog breed discrimination. VGG-16’s near zero silhouette at $K=5$ and negative at all higher K values on Oxford Cats indicates the feature space provides essentially no structured cat breed separation.

Limitations. Several limitations constrain the conclusions of this work. First, K-Means assumes spherical and approximately equal-size clusters — an assumption that is

visibly violated by the oversized clusters observed across multiple experiments. Breeds that share global visual attributes such as coat color or body shape are pulled into the same cluster, inflating its size and reducing ARI below what a more flexible clustering algorithm might achieve. Methods such as Gaussian Mixture Models or HDBSCAN, which do not impose spherical geometry, could better handle these irregular cluster boundaries.

Second, PCA dimensionality reduction is applied with a fixed component count rather than a principled variance threshold. VGG-16 features retained only 55.13% of variance at 100 components, meaning nearly half the representational signal was discarded before clustering even began. A higher component count or a non-linear alternative such as UMAP may preserve more discriminative structure and improve results.

Third, the pipeline has no robustness to nonconventional standard lighting, as the controlled brightness experiments demonstrate. Dog clustering ARI drops by up to 85% under very dark conditions which would be a critical failure mode for any real world deployment where lighting cannot be controlled. Brightness normalization or augmentation during feature extraction would likely mitigate this substantially.

Finally, all models were used as frozen pretrained extractors with no task specific adaptation. Fine tuning even a small portion of the network on a breed-labeled dataset would likely close the gap between CNN and ViT performance and improve absolute results across all conditions. SAM based background removal, which required 74.3 minutes on a T4 GPU, degraded performance in every condition tested, making it impractical for this pipeline in its current form.

5. Conclusion

This project set out to investigate whether pretrained convolutional neural networks could recover meaningful fine-grained breed structure from dog and cat images without any label supervision at clustering time. We evaluated two CNN architectures — ResNet-18 and VGG-16 — across four datasets spanning coarse to fine-grained difficulty: CIFAR-10 dogs, Imagewoof (10 breeds), Oxford Dogs (25 breeds), and Oxford Cats (12 breeds). Beyond the baseline pipeline, we conducted three robustness experiments examining SAM-based background removal, controlled brightness degradation, and an architectural comparison extending to Vision Transformers.

The first and most fundamental finding is that image resolution is the hard prerequisite for this task. Both ResNet-18 and VGG-16 fail completely on CIFAR-10’s native 32×32 images, producing silhouette scores near zero and near-random cluster distributions. The failure is not algorithmic — agglomerative clustering on VGG-16 CIFAR-10

features reaches a silhouette of 0.42, revealing that some geometric structure exists but is too non-spherical for K-Means to exploit. The deeper issue is that 32×32 pixels destroy the spatial detail that CNN convolutional filters require to extract breed-discriminative texture and morphological features. Upscaling to 224×224 cannot recover information that was never present. This result establishes a practical lower bound for the approach: any dataset below roughly 100 pixels on its shortest side should be expected to fail this pipeline.

At higher resolution, both CNNs recover meaningful breed structure. ResNet-18 achieves ARI 0.581 and average purity 77.3% on Imagewoof, and ARI 0.530 with purity 90.2% on Oxford Dogs — a particularly strong result where 22 of 25 breeds exceed 80% purity and cluster sharing is almost entirely biologically meaningful. VGG-16 achieves comparable performance with ARI 0.571 on Imagewoof and ARI 0.493 on Oxford Dogs. The comparison between these two models is the central empirical surprise of this work: VGG-16 produces feature vectors that are $8 \times$ larger than ResNet-18 (4,096 vs. 512 dimensions) and was expected to outperform on the basis of richer representation, yet neither model consistently dominates the other. ResNet-18 wins on Imagewoof and Oxford Dogs while VGG-16 edges ahead on Oxford Cats. This result points to a shared architectural constraint — global average pooling — that dilutes breed-discriminative foreground features by averaging them equally with background pixels, regardless of how many dimensions are produced downstream. Increasing feature dimensionality does not compensate for information lost at the pooling stage.

The SAM background removal experiment was designed to test whether explicitly isolating the foreground animal before feature extraction could overcome this limitation. It did not. The baseline without any segmentation outperformed both SAM variants — black fill and white fill — across all three datasets and both models, in all 18 dataset-pipeline comparisons without exception. The degradation stems from two compounding causes: imperfect SAM masks corrupting features for off-center or unusually posed animals, and the incompatibility of flat artificial fills with ImageNet-pretrained feature statistics. A particularly informative color-specific failure was observed: white fill severely degraded light-coated breeds such as Samoyed and Shih-Tzu because the animal’s coat became visually indistinguishable from the white background even when the segmentation mask was accurate.

The lighting experiment revealed that the pipeline’s reliance on color and brightness information is substantial and asymmetric across species. Dog ARI dropped 40–85% under dark and very dark conditions across both models, while cat ARI dropped only 23–69%. This asymmetry reflects the nature of the discriminative features each species relies on:

dog breeds are largely differentiated by coat color, markings, and color patterns, which are destroyed by brightness reduction, while cat breeds depend more on body shape and broad coat texture, which are more robust to illumination changes. Both CNN models exhibited this asymmetry similarly, suggesting it is a property of the task and the ImageNet feature space rather than of any specific architecture.

The architecture comparison provided the strongest positive result of the project. ViT-B/16 achieves ARI 0.796 and NMI 0.925 on Oxford Dogs — a 50% improvement over ResNet-18 and 62% over VGG-16. This result confirms that the global average pooling bottleneck identified in the CNN analysis is a genuine architectural limitation rather than a symptom of insufficient feature dimensionality. Vision Transformer self-attention implicitly learns to weight discriminative foreground regions more heavily than background regions, effectively solving the problem that SAM preprocessing attempted to address through segmentation. The ViT improvement required no additional preprocessing and no breed label information, arriving entirely from the architectural choice to replace pooling with attention.

Several limitations constrain the current conclusions. K-Means assumes spherical and approximately equal-size clusters, an assumption violated by the observed size imbalances across experiments. PCA dimensionality reduction is applied with a fixed component count rather than a principled variance threshold, and VGG-16 features retained only 55.13% of variance at 100 components, potentially discarding useful information. Purity analysis was only computed for a subset of conditions due to computational constraints. The pipeline has no robustness to non-standard lighting, as the brightness experiments demonstrate, which would be a significant limitation in real-world deployment.

Future directions include fine-tuning on a small breed-labeled dataset to adapt pretrained representations to the specific discriminative cues relevant to each species; replacing K-Means with density-based methods such as HDBSCAN or probabilistic Gaussian Mixture Models to better handle non-spherical cluster geometry; applying brightness and color augmentation during or after feature extraction to improve lighting robustness; and extending the architecture comparison to include domain-specific pretrained models such as those trained on veterinary imaging datasets. The results of this project suggest that for practical deployment, ViT based feature extraction should be the default choice for fine-grained unsupervised breed clustering, with CNN alternatives appropriate only when computational constraints prohibit the larger model size.

References

- [1] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. [1](#)

- [2] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. In *ICLR*, 2015. 1, 2
- [3] Alexey Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 1, 2
- [4] Alexander Kirillov et al. Segment anything. In *ICCV*, pages 4015–4026, 2023. 1, 3
- [5] Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *CVPR*, pages 3498–3505, 2012. 1
- [6] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. 1
- [7] Jeremy Howard et al. Imagewoof dataset. <https://github.com/fastai/imagenette>, 2020.
- [8] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *JMLR*, 12:2825–2830, 2011. 2
- [9] Adam Paszke et al. PyTorch: An imperative style, high-performance deep learning library. In *NeurIPS*, volume 32, 2019. 2
- [10] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *JMLR*, 9:2579–2605, 2008. 2
- [11] Eric Rhodes. Automatic dog breed identification. Technical report, 2020. 1